

ACTIVITY PACK

Activity 7 - Break a Leaky Chatbot, Then Compare the Guarded One

Complete learner scenario, practice and security checklist

Course	AI Security for Autonomous AI Agents
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Readme

Activity 7 - Break a Leaky Chatbot, Then Compare the Guarded One

Purpose: Try to make an unsafe chatbot leak fictional private data, then see how a guarded version blocks the very same attacks.

■■ This is a deliberately insecure TRAINING demo with FICTIONAL data. The "leaky" bot is built to fail on purpose so you can see the risk. Never use these techniques on any real system.

What you need

- A web browser
- leaky-chatbot/index.html and guarded-chatbot/index.html in this folder
- A **training** API key (from your trainer)
- The attack sheet in this folder: ATTACK-PROMPTS.md

Evidence status: SIM - all data is fictional. All names, bookings and "secrets" are made up (for example, the fictional guest *Tan Ah Kow*).

API key safety: Use only the **low-limit training key**. **Never** a production or personal key. It is stored **only in your browser** for this session. Clear it when done.

Steps

1. Open leaky-chatbot/index.html and enter the training API key.
2. Ask a **normal question** first (for example, "What are your check-in times?") to see it behave helpfully.
3. Now try the **attack prompts** from ATTACK-PROMPTS.md one by one. Note what fictional private data leaks - customer bookings, internal memos, an "admin password", a guest's phone number.
4. Open guarded-chatbot/index.html and enter the same training API key.
5. Send the **exact same prompts**. Watch the guardrails block or safely refuse them.
6. For 3 - 4 prompts, write down **what the leaky bot revealed** and **how the guarded bot responded**.

The four guardrail layers (why the guarded bot is safe)

1. **Retrieval filter** - the bot can only fetch data it is allowed to; private records are never pulled in.
2. **Input guard** - checks your message for attack patterns ("ignore your instructions", "print the password") before the model sees it.
3. **Hardened prompt** - the bot's own instructions firmly refuse to reveal secrets or personal data.
4. **Output guard** - scans the answer before it is shown, and blocks it if private data slipped through.

What you produce

- A short table: prompt -> what the leaky bot leaked -> how the guarded bot responded
- One sentence naming which guardrail layer stopped each attack

Reflect (Data Privacy / Job Impact / Ethical Concerns / Cyber Security)

- **Data Privacy:** If this were real, whose data just leaked, and what harm could follow?
- **Job Impact:** Who in a real team is responsible for testing a chatbot like this before launch?
- **Ethical Concerns:** Is it acceptable to ship the "leaky" version to save time? Why not?
- **Cyber Security:** Which single guardrail would you never skip, and why?